Physics-101, Mid-semester examination

Department of Physics

Shiv Nadar University (SNU)

Time: 90 Minutes Max. Marks: 25.

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1) (i) A car moves in a circular path of radius R. It starts from the rest while the speed increases at a constant rate α. Determine the angle between its velocity and acceleration vectors at time t. [2.5]

(ii) In the following figure (Fig .1), the hand is pulling the thread with acceleration a, Find the acceleration of Block using the constraint of inextensible rope and choosing proper coordinate system. [2.5]

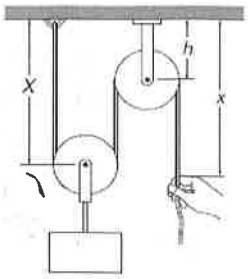


Fig. 1

(iii) A rubber ball of mass 0.2 kg falls to the floor. The ball hits with a speed of $8m/s$ and rebounds with approximately the same speed. High speed photographs show that the ball is in contact with the floor for $\Delta t = 10^{-3}\ s$. What is the average force exerted on the ball by the floor? [2.5]

(v) A ball of mass m is released from a point inside a frictionless bowl of radius R. When the ball makes an angle with respect to vertical say θ, as shown in Fig. 2 , during its periodic motion, Draw free body diagram and write equation of motion in the given quantities using polar coordinate system. [2.5]

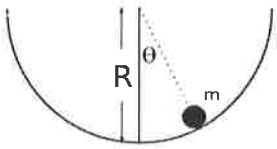


Fig.2

2) A disk rotates with constant angular velocity ω, as shown in Fig. 3. Two masses, m_A and m_B, slide without friction in a groove passing through the center of the disk. They are connected by a light string of length l, and are initially held in position by a catch, with mass m_A at distance r_A from the center. Neglect gravity. At $t = 0$ the catch is removed and the masses are free to slide.

(i) Draw free body diagram of each mass from the point of view of an observer on the ground. (Note, show only the forces in the free body diagram. Do not write the acceleration generated by these forces in the same diagram.) [1]

(ii) Write Newton's equation for each particle in polar coordinate system taking center of the circle as origin for the observer from the ground.

(Note, while writing the equations $\vec{F} = m\vec{a}$, do not mix left hand side and right hand side of the equations. The terms related to the force and acceleration should be expressed explicitly.) [1]

(iii) Draw free body diagram of each mass from the point of view of an observer in the rest frame of particle (rotating frame). (Note, show only the forces in the free body diagram. Do not write the acceleration generated by these forces in the same diagram.) [1]

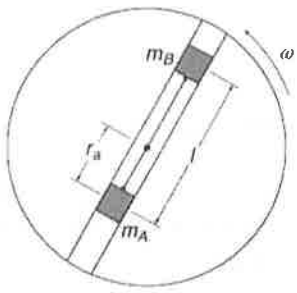


Fig. 3

(iv) Write Newton's equation for each particle in polar coordinate system taking center of the circle as origin for the observer in the rest frame of particle (rotating frame).(Note, while writing the equations $\vec{F} = m\vec{a}$, do not mix left hand side and right hand side of the equations. The terms related to the force and acceleration should be expressed explicitly.) [1]

(v) Find $\ddot{r}_A$ immediately after the catch is removed, in terms of m_A, m_B, l, r_A, and ω. [1]

3) A block rests on a wedge inclined at angle θ. The coefficient of static friction between the block and wedge is μ. In this question do the analysis from the ground observer only.
(*a*) Find the maximum value of θ for the block to remain motionless on the wedge when the wedge is fixed in position. [1]
(*b*) The wedge is given horizontal acceleration *a*, as shown. Assuming that $\tan\theta > \mu$, find the minimum acceleration for the block to remain on the wedge without sliding. [3]
(*c*) Repeat part (*b*), but find the maximum value of the acceleration [1]

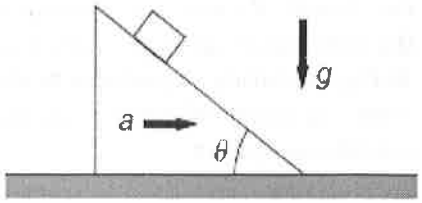
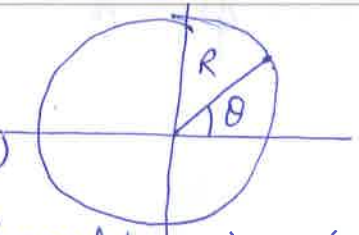


Fig. 4

4) A cannon fires successively two shells with velocity v_0 = 250 m/s; the first at the angle θ_1 = 60° and the second at the angle θ_2 = 45° to the horizontal, the azimuth being the same. Neglecting the air drag, find the time interval between firings leading to the collision of the shells. [5]

1(i) In polar coordinate system



$\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$ (velocity) —(i)

$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}$ (acceleration) —(ii)

for this case $\vec{v} = R\dot{\theta}\hat{\theta}$; $\dot{r} = 0$; particle is moving on circle

Speed $= |\vec{v}| = R|\dot{\theta}| = v = R\dot{\theta}$ —— (iii) [0.5]

Given, speed increases with constant rate $\Rightarrow \frac{dv}{dt} = \alpha \Rightarrow R\ddot{\theta} = \alpha$ —(IV)

particle start from rest $\Rightarrow \dot{\theta}\Big|_{t=0} = 0 \Rightarrow \dot{\theta} = \frac{\alpha}{R}t$ —— (V)

Now $\vec{v} = R\dot{\theta}\hat{\theta} = R\frac{\alpha}{R}t\hat{\theta} = \alpha t\hat{\theta}$ —— (VI)

$$\vec{a} = -R\left(\frac{\alpha}{R}t\right)^2\hat{r} + \left(R\frac{\alpha}{R} + 2\times 0\right)\hat{\theta}$$

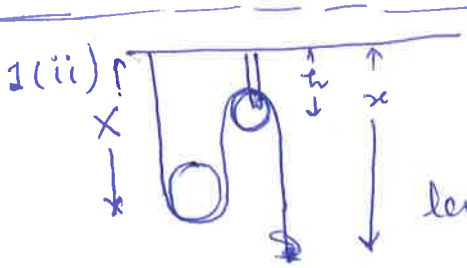
$\vec{a} = -\frac{\alpha^2 t^2}{R}\hat{r} + \alpha\hat{\theta}$ —— (VII) [0.5]

Now, $\vec{v}\cdot\vec{a} = |\vec{v}||\vec{a}|\cos(\text{angle}) = \alpha^2 t$ —— (VIII)

$$\text{angle} = \cos^{-1}\left[\frac{\alpha^2 t}{|v||a|}\right] = \cos^{-1}\left[\frac{\alpha^2 t}{\alpha t\sqrt{\left(\frac{\alpha^2 t^2}{R}\right)^2 + \alpha^2}}\right] \quad -(IX)$$

[0.5]

1(ii)



Assuming top line or support as reference point

down ward direction +ve

length of Rope l ; $l = X + X - h + x - h = 2X + x - 2h$

$\Rightarrow 2X + x = l + 2h$ (constant) $\Rightarrow 2\ddot{X} + \ddot{x} = 0$ [0.5]

$\Rightarrow \ddot{X} = -\frac{\ddot{x}}{2}$; given that $\ddot{x} = a \Rightarrow \ddot{X} = -\frac{a}{2}$ [0.5]

1 (iii)

From Newton's law

$$\vec{F} = \frac{\Delta \vec{P}}{\Delta t}$$

[0.5]

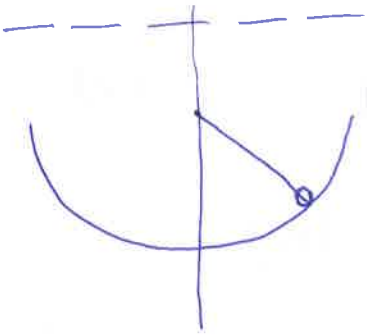
$$\vec{F} = \vec{F}_{wall} + F_{gravity} = \frac{\Delta \vec{P}}{\Delta t}$$

Neglecting gravity because it is very small in comparison to $\vec{F}_{wall}$

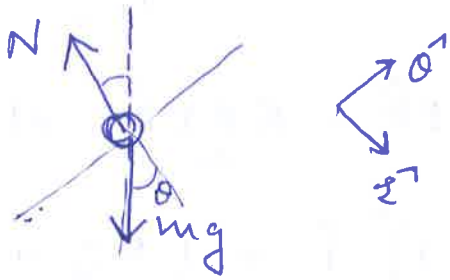
$$\vec{F}_{wall} = \frac{\Delta \vec{P}}{\Delta t} = \frac{P_f - P_i}{\Delta t} = \frac{mv\hat{j} - (mv(-\hat{j}))}{\Delta t} = \frac{2mv\hat{j}}{\Delta t}$$ [1.5]

$$\vec{F} = \frac{0.2 \times 2 \times 8}{10^{-3}}\hat{j} = 3.2 \times 10^{3}\, N\hat{j}$$ [0.5]

1(iv)



Free body diagram



[0.5]

For polar coordinate system

$$\vec{F} = F_r\hat{r} + F_\theta\hat{\theta} = (-N + mg\cos\theta)\hat{r} + (-mg\sin\theta)\hat{\theta}$$ [0.5]

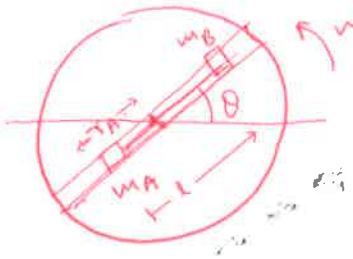
Equation of motion

$$\vec{F} = m\vec{a} = m(a_r\hat{r} + a_\theta\hat{\theta}) = m(\ddot{r} - r\dot{\theta}^2)\hat{r} + m(r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}$$ [0.5]

$$(-N + mg\cos\theta)\hat{r} + (-mg\sin\theta)\hat{\theta} = m(\ddot{r} - r\dot{\theta}^2)\hat{r} + m(r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}$$
$$= -mR\dot{\theta}^2\hat{r} + mR\ddot{\theta}\hat{\theta}$$

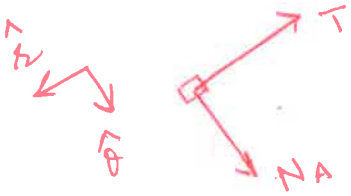
$$\Rightarrow (-N + mg\cos\theta) = -mR\dot{\theta}^2 \quad \text{——— (i)}$$ [0.5]

$$-mg\sin\theta = mR\ddot{\theta} \quad \text{——— (ii)}$$ [0.5]



2.(i) Free body diagram

mass A



[0.5]

mass B

[0.5]

2(ii) Newton's Equation

$$\vec{F} = F_r\hat{r} + F_\theta\hat{\theta} = m(a_r\hat{r} + a_\theta\hat{\theta})$$

For particle A

$$-T\hat{r} + N_A\hat{\theta} = m_A(\ddot{r}_A - r_A\dot{\theta}^2)\hat{r} + m_A(r_A\ddot{\theta} + 2\dot{r}_A\dot{\theta})\hat{\theta}$$

at The moment catch is released $(t=0)$ $\dot{r}_A = 0$ and $\ddot{\theta} = 0$, $\dot{\theta} = \omega = \text{constant}$

$$\Rightarrow -T\hat{r} + N_A\hat{\theta} = m_A(\ddot{r}_A - r_A\omega^2)\hat{r} + m_A 0\,\hat{\theta}$$

comparing $-T = m_A(\ddot{r}_A - r_A\omega^2)$, $N_A = 0$ [0.5]

For particle B

$$-T\hat{r} + N_B\hat{\theta} = m_B(\ddot{r}_B - r_B\dot{\theta}^2)\hat{r} + m_B(r_B\ddot{\theta} + 2\dot{r}_B\dot{\theta})\hat{\theta}$$

$$= m_B(\ddot{r}_B - r_B\omega^2)\hat{r} + 0\,\hat{\theta}$$

$$\Rightarrow -T = m_B(\ddot{r}_B - r_B\omega^2), \quad N_B = 0$$

[0.5]

2 (iii) The rotating frame of reference is an noninertial frame hence in this frame there is an acceleration giving pseudo force (let me denote this new frame by prime)

Free body diagram for A

[0.5]

Freebody diagram for B

[0.5]

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2 (iv) $\vec{F}' = F_r' \hat{r}' + F_\theta' \hat{\theta} = m(a_r' \hat{r}' + a_\theta' \hat{\theta}')$

In this frame particle is not rotating (for all time) $\Rightarrow \dot{\theta}' = 0$

$$a_r' = \ddot{r}' - r'\dot{\theta}'^2 = \ddot{r}'$$

$$a_\theta' = r\ddot{\theta}' + 2\dot{r}'\dot{\theta}' = 0$$

$\Rightarrow$ From free body diagram for particle A

$$(-T + m_A r_A \omega^2)\hat{r}' + N_A \hat{\theta}' = m_A \ddot{r}_A \hat{r}' + 0\,\hat{\theta}' \quad [0.5]$$

Similarly for particle B

$$(-T + m_B r_B \omega^2)\hat{r}' + N_B \hat{\theta}' = m_B \ddot{r}_B \hat{r}' + 0\,\hat{\theta}' \quad [0.5]$$

$$\Rightarrow \quad -T + m_A r_A \omega^2 = m_A \ddot{r}_A, \quad N_A = 0 \quad \text{--- (i)}$$

$$-T + m_B r_B \omega^2 = m_B \ddot{r}_B, \quad N_B = 0 \quad \text{--- (ii)}$$

2 (V) From (i) & (ii)

adding (i) & (ii) after multiply m_B & m_A respectively

$$m_B(-T + m_A r_A \omega^2) = m_B m_A \ddot{r}_A$$

$$m_A(-T + m_B r_B \omega^2) = m_A m_B \ddot{r}_B$$

$$\Rightarrow \quad -T(m_A + m_B) + m_A m_B (r_A + r_B)\omega^2 = m_A m_B (\ddot{r}_A + \ddot{r}_B)$$

We know $r_A + r_B = l$ = constant

$$\Rightarrow \quad \ddot{r}_A + \ddot{r}_B = 0$$

$$\Rightarrow \quad -T(m_A + m_B) + m_A m_B l \omega^2 = 0$$

$$T = \frac{m_A m_B l \omega^2}{(m_A + m_B)}$$

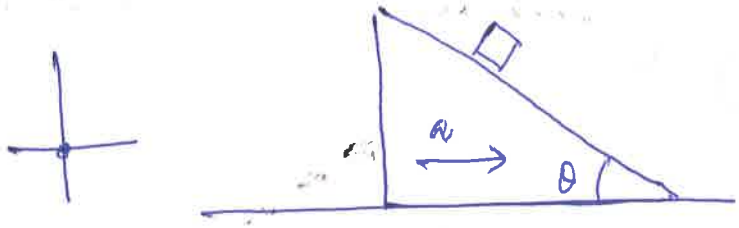
putting the value of T in (i)

$$m_A \ddot{r}_A = -\frac{m_A m_B l \omega^2}{(m_A + m_B)} + m_A r_A \omega^2$$

$$\boxed{\ddot{r}_A = -\frac{m_B l \omega^2}{(m_A + m_B)} + r_A \omega^2}$$

[1]

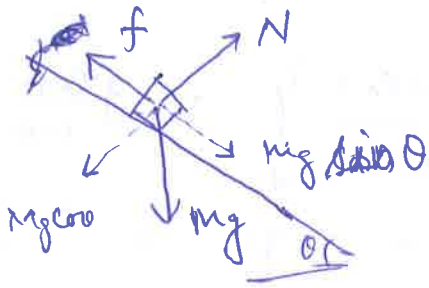
3)



From the point of view of ground observer.

~~given that~~

(a)



For balancing situation.

$f = Mg\sin\theta$

$N = Mg\cos\theta$

as we increase θ, $mg\sin\theta$ increases hence f increases [0.5]

$f \le \mu N$; $f_{max} = \mu N$

$\Rightarrow \quad mg\sin\theta_{max} = f_{max}$ for No slipping

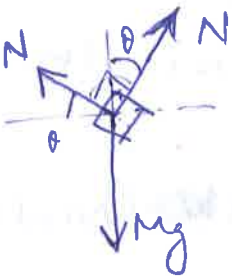
$= \mu N$

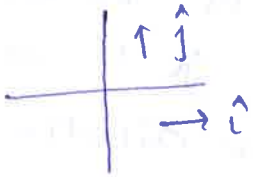
$Mg\sin\theta_{max} = \mu Mg\cos\theta_{max} \quad \Rightarrow \quad \mu = \tan\theta_{max}$

$\boxed{\theta_{max} = \tan^{-1}\mu}$ [0.5]

(b) Given that $\theta > \theta_{max}$ so the tendency of the block is to slip downward along the slope. hence the friction will be upward along the slope [1]

free body diagram $f = \mu N$





$\vec{N} = N\sin\theta\,\hat{i} + N\cos\theta\,\hat{j}$

$\vec{f} = \overrightarrow{(\mu N)} = -\mu N\cos\theta\,\hat{i} + \mu N\sin\theta\,\hat{j}$

$\vec{Mg} = -Mg\,\hat{j}$

$\vec{F} = \vec{N} + \overrightarrow{(\mu N)} + \vec{Mg} = m\vec{a}$

$= (N\sin\theta - \mu N\cos\theta)\,\hat{i} + (N\cos\theta + \mu N\sin\theta - Mg)\,\hat{j} = M\vec{a}$

$= Ma\,\hat{i}$ [1]

$\Rightarrow$

$$N\sin\theta - \mu N\cos\theta = Ma$$

$$N\cos\theta + \mu N\sin\theta - Mg = 0$$

$$\frac{Ma}{Mg} = \frac{N(\sin\theta - \mu\cos\theta)}{N(\cos\theta + \mu\sin\theta)}$$

$$a = g\left\{\frac{\sin\theta - \mu\cos\theta}{\cos\theta + \mu\sin\theta}\right\} = g\frac{\cos\theta}{\cos\theta}\left(\frac{\tan\theta - \mu}{1+\mu\tan\theta}\right)$$

This is the minimum acceleration required. $= g\left(\frac{\tan\theta - \mu}{1+\mu\tan\theta}\right)$ [1]

as If $\theta = \theta_{max}$. a should be zero

$$a = g\left(\frac{\tan\theta_{max} - \mu}{1+\mu\tan\theta_{max}}\right) = g\frac{(\mu-\mu)}{(1+\mu^2} = 0$$

But for $\theta > \theta_{max}$ $\tan\theta > \tan\theta_{max} \Rightarrow a > 0$

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8 (c) for maximum value of 'a'

as we know to increas the acceleration N must increas.

so $N\cos\theta + \mu N\sin\theta > Mg$, so μN will be reduced and it will go to f [0.5]

$\Rightarrow$ $N\cos\theta + f\sin\theta = Mg$

$N\sin\theta - f\cos\theta = Ma$

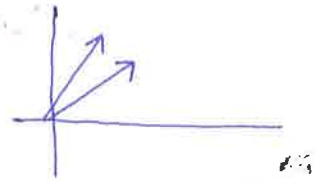
with increase of N to balance the equation f will keep reducing untill it go to zero then reverse and finall maximum value in reverse direction. Hence Now [0.5]

New equation.

$$\vec{F} = (N\sin\theta + \mu N\cos\theta)\hat{i} + (N\cos\theta - \mu N\sin\theta - Mg)\hat{j} = Ma\hat{i}$$

$$\Rightarrow \begin{aligned} Ma &= N(\sin\theta + \mu\cos\theta) \\ Mg &= N(\cos\theta - \mu\sin\theta) \end{aligned} \Rightarrow a = g\left(\frac{\tan\theta + \mu}{1 - \mu\tan\theta}\right)$$

4)



Let the interval between firing be Δt and it hit at time ~~before~~ t after the firing of second particle

when they hit there coordinate must be same

so X coordinate of the particle a (first)

$$X_a = V_0 \cos 60^\circ \times (t + \Delta t)$$

for X coordinate of the particle b (second)

$$X_b = V_0 \cos 45^\circ \times t$$

$$X_a = X_b \Rightarrow \quad \cancel{V_0} \cos 60^\circ (t + \Delta t) = \cancel{V_0} \cos 45^\circ \times t$$

$$\frac{1}{2}(t + \Delta t) = \frac{1}{\sqrt{2}} t$$

$$\Rightarrow \quad t = \frac{\Delta t}{(\sqrt{2} - 1)} \quad \text{— (1)}$$

[0.5]

[1.5]

for Y coordinate

$$Y_a = V_0 \sin 60^\circ (t + \Delta t) - \frac{1}{2} g (t + \Delta t)^2$$

$$Y_b = V_0 \sin 45^\circ t - \frac{1}{2} g t^2$$

$$Y_a = Y_b \Rightarrow \quad V_0 \sin 60^\circ (\sqrt{2} t) - \frac{1}{2} g (\sqrt{2} t)^2 = V_0 \sin 45^\circ t - \frac{1}{2} g t^2$$

[2]

$$V_0 t \left(\frac{\sqrt{6}}{2} - \frac{1}{\sqrt{2}}\right) = \left(1 - \frac{1}{2}\right) g t^2$$

$$V_0 \frac{(\sqrt{3} - 1)}{\sqrt{2}} = \frac{1}{2} g t \Rightarrow t = \frac{V_0 \sqrt{2} (\sqrt{3} - 1)}{g}$$

from (1) $\quad \Delta t = (\sqrt{2} - 1) \frac{V_0 \sqrt{2} (\sqrt{3} - 1)}{g}$

[1]